B.Sc. Engg. CSE 5th Semester 10 April 2026 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2024-2025**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4503: Microprocessors and Assembly Language

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Classify different types of Interrupts. Explain the differences between different types of Hardware Interrupts. 2 + 5 (CO1) (PO1)

   b) What is Duty Cycle? Find out the relation between Clock State, Bus/Machine Cycle, and Instruction Cycle. 2 + 6 (CO1) (PO1)

   c) Compare and contrast between the control signal operations of Maximum and Minimum modes of 8086 microprocessor. 10 (CO4) (PO2)

2. a) Draw the details of WRITE Bus timing diagram showing all the necessary/required signals of 8086. You should consider that the write operation is to be made to an output device port address of `10101h`. Consider that there is 4 `WAIT` states, interrupt flag is SET, and DS register is in use for data accessing. 7 (CO4) (PO2)

   b) Suppose, while debugging an assembly language program the values of the registers are: `Flag=FEB9h`, `IP=0102h`, `CS=0700h`, `SP=FFFAh`. Now, if `INT 01000001B` is requested, derive the memory addresses from where the new IP and CS can be retrieved; Also show the step-by-step changes in memory contents of stack segment along with corresponding SP values while handling the interrupt by the 8086 microprocessor. 8 (CO4) (PO4)

   c) What is MSW register? Show the comparative advancements of Status/Flag registers of different microprocessors starting from 8085 to new age processors. 2 + 8 (CO1) (PO2)

3. a) Differentiate between Protected mode and VM8086 mode. 7 (CO2) (PO2)

   b) Draw the structures of memory banks of 80286, 80386, and Pentium. Also, mention how the address pins are used to identify the memory banks. 8 (CO3) (PO1)

   c) What is FPU? Explain the pipelining concept of Pentium Processors using for U-pipeline, V-pipeline and FPU pipeline stages. 2 + 8 (CO3) (PO1)

4. a) Explain the different categories of memory segmentation model. 7 (CO3) (PO3)

   b) What is Paging? Describe the process of calculation of linear address while paging is enabled and disabled. 2 + 6 (CO2) (PO1)

c) Suppose, you are given a memory address of 4GB (with addresses 00000000h to FFFFFFFFh). Using a 16-bit segment register, clearly explain the process to define a global descriptor table along with local descriptor table. 10 (CO4) (PO2)

5. a) What is Multi-core processor and how does multi-core processor work? 7 (CO3) (PO1)

b) Explain the concepts of Thread, Hyper-thread, and Multi-thread with respect to multi-core processor system. 8 (CO3) (PO1)

c) Analyze the performance and principle of "Single/Multiple Instruction Stream and Single/-Multiple Data Stream" for designing single/multi-core processors with appropriate examples. 10 (CO3) (PO3)

6. a) Discuss the differences between CPU, Microcontroller, and GPU. Your answer should address their architectural design, applications, operations, and features. 10 (CO1) (PO1)

b) Write short notes on the followings using appropriate example/figure: 5×3 (CO1) (PO1)

i. GPU Pipeline

ii. AVR Microcontrollers

iii. ATmega16 Status Register